INCH - POUND

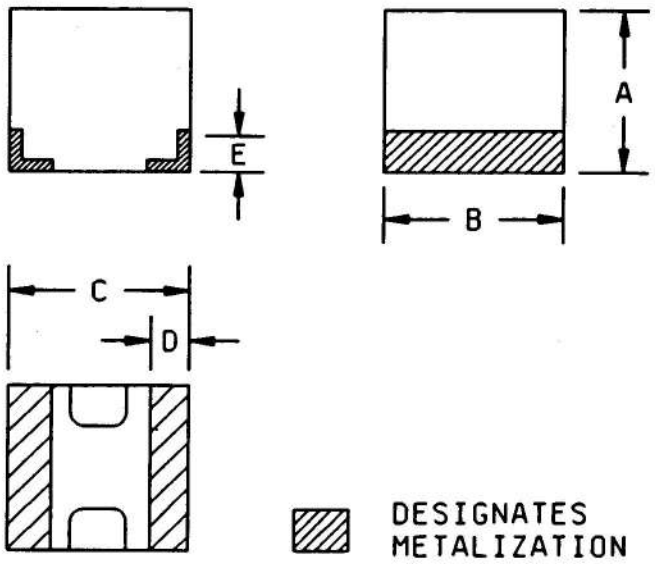
MIL-PRF-83446/13C
w/AMENDMENT 1
7 April 2019
SUPERSEDING
MIL-PRF-83446/13C
1 April 2014

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-83446.



	INCHES	mm
A	.100 ±.005	2.54 ±0.13
B	.100 ±.005	2.54 ±0.13
C	.100 ±.005	2.54 ±0.13
D	.025 ±.005	0.64 ±0.13
E	.015 Minimum	0.38 Minimum

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Fixed, chip coil.

AMSC N/A

FSC 5950

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Diallyl phthalate.

Weight: 0.02 ounce, maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: MIL-STD-202-301, test voltage 700 volts rms.

Barometric pressure: MIL-STD-202-105, test condition C, (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Part or Identifying Number (PIN): The part number shall be in the following form.

M83446/13-	04	B
	Sequentially assigned dash Numbers (see table I and II)	Termination finish (see MIL-PRF-83446)

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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number	Inductance ±10% (μH)	Q (min)	Q (typ)	Test Frequency (MHz) 1/	Self resonant frequency (min) MHz	DC resistance (max) ohms	Rated current (max) mA 2/	Incremental current mA 3/
01	0.010	45	60	200	1200	0.020	750	----
02	0.012	45	60	200	1200	0.025	750	----
03	0.015	45	60	200	1200	0.035	750	----
04	0.018	45	60	200	1200	0.040	750	----
05	0.022	45	60	200	1000	0.045	750	----
06	0.027	45	55	200	1000	0.050	750	----
07	0.033	45	55	200	1000	0.055	640	----
08	0.039	45	55	200	1000	0.060	600	----
09	0.047	45	50	100	1000	0.065	550	----
10	0.056	45	50	100	100	0.060	520	----
11	0.068	40	45	100	800	0.060	480	----
12	0.082	35	40	100	800	0.080	470	----
13	0.100	35	40	25.0	750	0.160	460	----
14	0.120	35	40	25.0	700	0.170	455	----
15	0.150	35	40	25.0	650	0.190	450	----
16	0.180	35	40	25.0	600	0.200	430	----
17	0.220	35	40	25.0	550	0.300	350	----
18	0.270	35	40	25.0	480	0.350	310	----
19	0.330	35	40	25.0	400	0.720	280	----
20	0.390	35	40	25.0	380	0.80	250	----
21	0.470	35	40	25.0	350	1.26	230	----
22	0.560	35	40	25.0	330	1.30	220	----
23	0.680	35	40	25.0	300	1.40	210	----
24	0.820	35	40	25.0	250	1.50	200	----
25	1.0	35	40	25.0	220	1.60	200	----
26	1.2	32	40	7.90	200	1.70	200	----
27	1.5	32	40	7.90	175	2.00	190	----
28	1.8	32	40	7.90	150	2.00	185	----

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers – Continued.

Dash number	Inductance ±10% (μH)	Q (min)	Q (typ)	Test Frequency (MHz) 1/	Self resonant frequency (min) MHz	DC resistance (max) ohms	Rated current (max) mA 2/	Incremental current mA 3/
29	2.2	32	40	7.90	95	2.0	180	----
30	2.7	30	35	7.90	85	2.00	175	----
31	3.3	30	35	7.90	70	2.00	170	----
32	3.9	30	35	7.90	60	2.10	165	----
33	4.7	30	35	7.90	50	2.30	165	----
34	5.6	30	35	7.90	40	2.50	165	----
35	6.8	30	35	7.90	30	2.50	165	----
36	8.2	27	32	7.90	22	2.50	120	----
37	10	35	40	7.90	20	2.90	105	58
38	12	35	40	2.50	15	3.00	91	53
39	15	35	40	2.50	13	3.00	87	50
40	18	35	40	2.50	12	3.20	81	46
41	22	35	40	2.50	10	4.00	77	37
42	27	35	40	2.50	9.0	4.00	73	34
43	33	35	40	2.50	9.0	4.50	69	32
44	39	35	40	2.50	6.5	5.00	65	29
45	47	35	40	2.50	6.2	5.70	63	29
46	56	35	40	2.50	6.0	6.00	60	28
47	68	35	40	2.50	5.8	6.70	57	26
48	82	30	40	2.50	5.2	7.50	55	23
49	100	30	35	2.50	4.0	13.0	50	20
50	120	28	30	.790	4.0	14.0	40	20
51	150	28	30	.790	3.9	18.0	36	20
52	180	28	30	.790	3.5	20.0	33	19
53	220	28	30	.790	3.2	21.0	28	18
54	270	20	25	.790	3.1	25.0	25	17
55	330	20	25	.790	2.8	37.0	20	16
56	390	20	25	.790	2.5	39.0	17	16
57	470	20	25	.790	2.3	45.0	15	15
58	560	20	25	.790	2.2	57.0	13	14
59	680	20	25	.790	2.1	62.0	12	13
60	820	18	20	.790	2.1	78.0	11	11
61	1000	10	15	.790	1.1	96.0	10	10

1/ Test frequency range 0.25 through 25 MHz: For electrical characteristics measurements, use the HP260-A Q Meter, HP4342A, HP250B RX Meter or equivalent with TF-405 test fixture, or equivalent (see figure 2). Fixture inductance (approximately 0.038 μH) and residual Q-meter inductance (approximately 0.01 μH) should be subtracted from indicated inductance.

Test frequency range 25 through 200 MHz: For electrical characteristics measurements, use the HP190-A Q Meter, HP260-A Q Meter, HP4342A, HP250B RX Meter or equivalent with TF-405 test fixture, or equivalent (see figure 2). Fixture inductance (approximately 0.009 μH) should be subtracted from indicated inductance.

2/ Maximum current allowed is not to exceed the specified temperature rise.

3/ Applicable to ferrite core only.

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurements			
	Inductance (Percent)	DC resistance	Self-resonant frequency (Percent)	Q (Percent)
Qualification inspection				
Group II	±5	±(3% +.001 ohm)	-8	-10
Group IV	±5	±(2% +.001 ohm)	-10	-10
Group V	±2	---	---	-10
Conformance inspection group C				
Subgroup II	±5	±(3% +.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +.001 ohm)	-8	-10

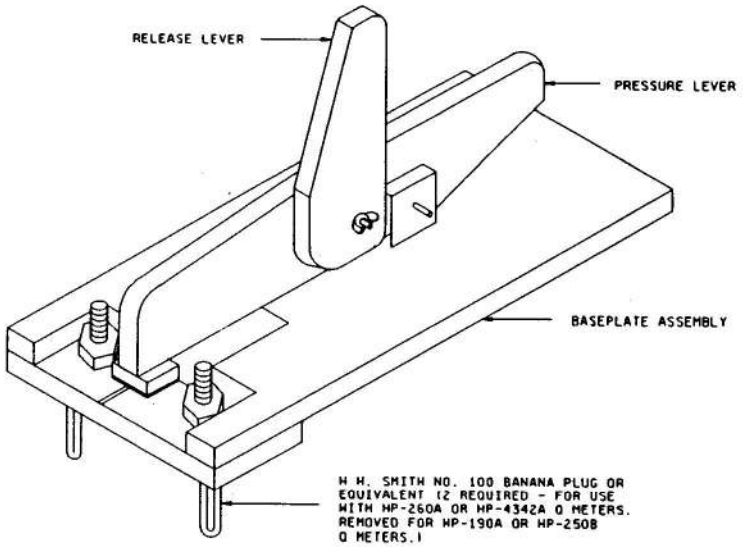


FIGURE 2. Typical chip coil test fixture (TF-405, or equivalent).

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-83446, this document references the following:

MIL-STD-202-105 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA - CC

Review activities:
Army – MI
Navy – AS, OS
Air Force - 19

Preparing activity:
DLA – CC

(Project 5950-2018-078)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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